Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

ITEP 414

System Administration and Maintenance

Week 3 Portfolio Project

Enterprise Server Deployment

and

Operating System Installation

MATIENZO, JOHN LESTER D.

BSIT 4C

August 15, 2026

Installation Guide of Windows Server Evaluation using Oracle VM VirtualBox

Project Overview

This activity focuses on the installation and basic configuration of Windows Server Evaluation using Oracle VM VirtualBox. The activity was completed as part of the ITEP 414 - System Administration and Maintenance course to provide hands-on experience in installing a server operating system, configuring its basic settings, and verifying that the installed system is functioning correctly.

The installation was performed using virtualization software, allowing Windows Server to be installed and tested without affecting the main operating system of the computer. The documentation covers the virtual machine specifications, installation procedure, initial configuration, verification commands, BIOS and UEFI comparison, boot process, encountered problems, implemented solutions, and lessons learned.

Objectives

1. To install Windows Server Evaluation using Oracle VM VirtualBox.
2. To understand the basic requirements needed for a Windows Server installation.
3. To configure the virtual machine with appropriate CPU, RAM, storage, and network settings.
4. To understand the difference between BIOS and UEFI boot methods.
5. To verify the successful installation and operation of Windows Server using system commands.
6. To document the complete installation process using screenshots.
7. To identify installation problems and apply appropriate solutions.
8. To develop practical skills in server operating system installation and basic system administration.

Software Requirements

Software	Purpose
Oracle VM VirtualBox	Creates and manages the virtual machine
Windows Server Evaluation ISO	Operating system installation media
Windows 10/11 Host OS	Main operating system
Web Browser	Downloading the Windows Server ISO
Microsoft Windows Server	Server operating system being installed

Virtual Machine Specifications

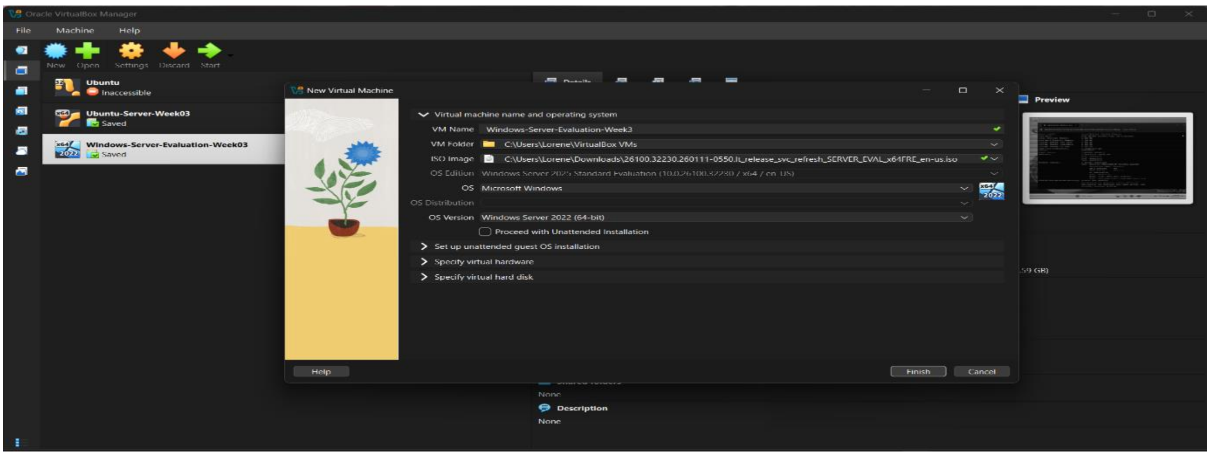
Component	Specification
Virtualization Software	Oracle VirtualBox
Operating System	Windows Server Evaluation
Architecture	64-bit
RAM	4 GB or more
CPU	2 processors
Storage	50–60 GB
Network Adapter	NAT
Boot Mode	BIOS/UEFI depending on configuration
Installation Media	Windows Server Evaluation ISO

Step-by-Step Installation Procedure

Step 1 — Create the Virtual Machine

Open VirtualBox and create a new VM.
Typical process:
VirtualBox → New
Enter:

- Name: Windows Server
- Type: Microsoft Windows
- Version: Windows 2022/2025 (64-bit) depending on your available VirtualBox option
- RAM: your assigned amount
- CPU: your assigned amount
- Virtual hard disk: your assigned storage

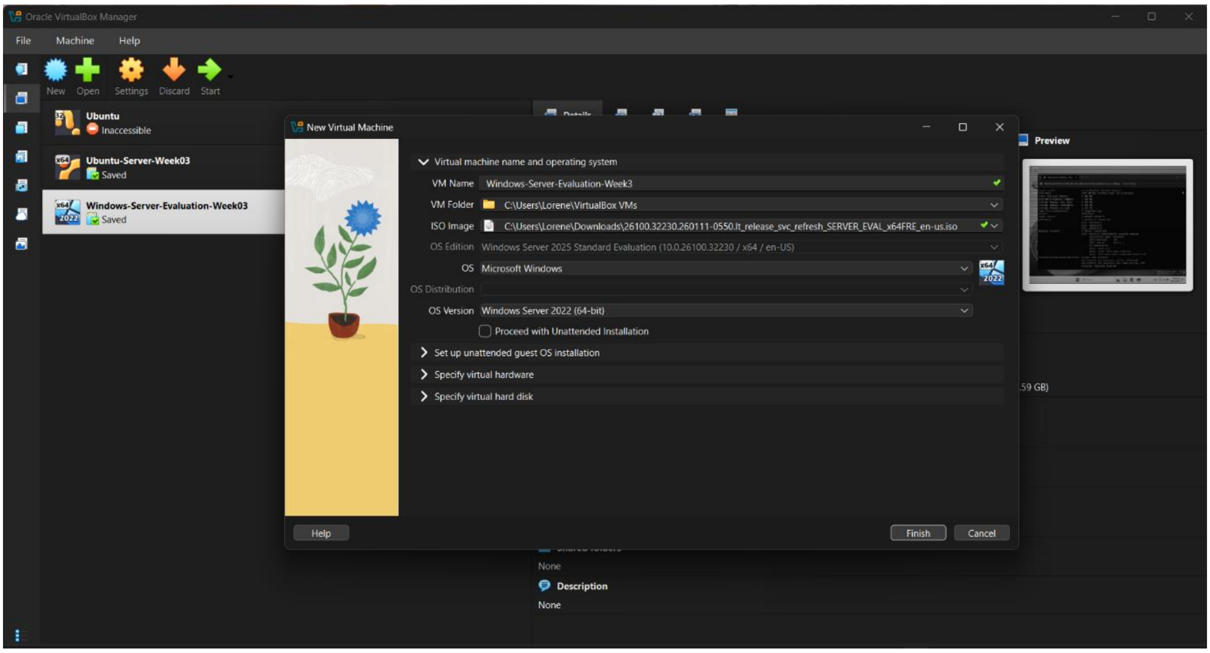


Step 2 — Attach the Windows Server ISO then Start the Virtual Machine

Go to:

Settings → Storage

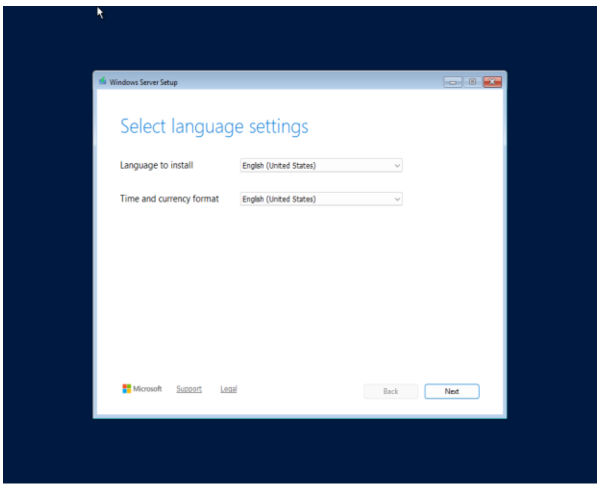
Select the virtual optical drive and attach your Windows Server Evaluation ISO.

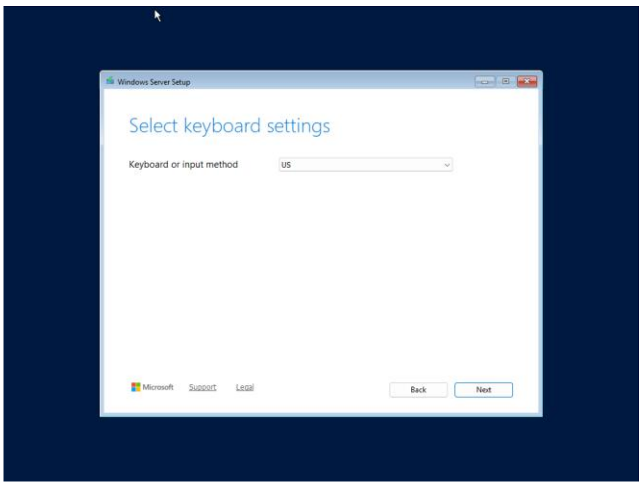


Step 3 — Select Language and Keyboard

Windows Setup will display options such as:

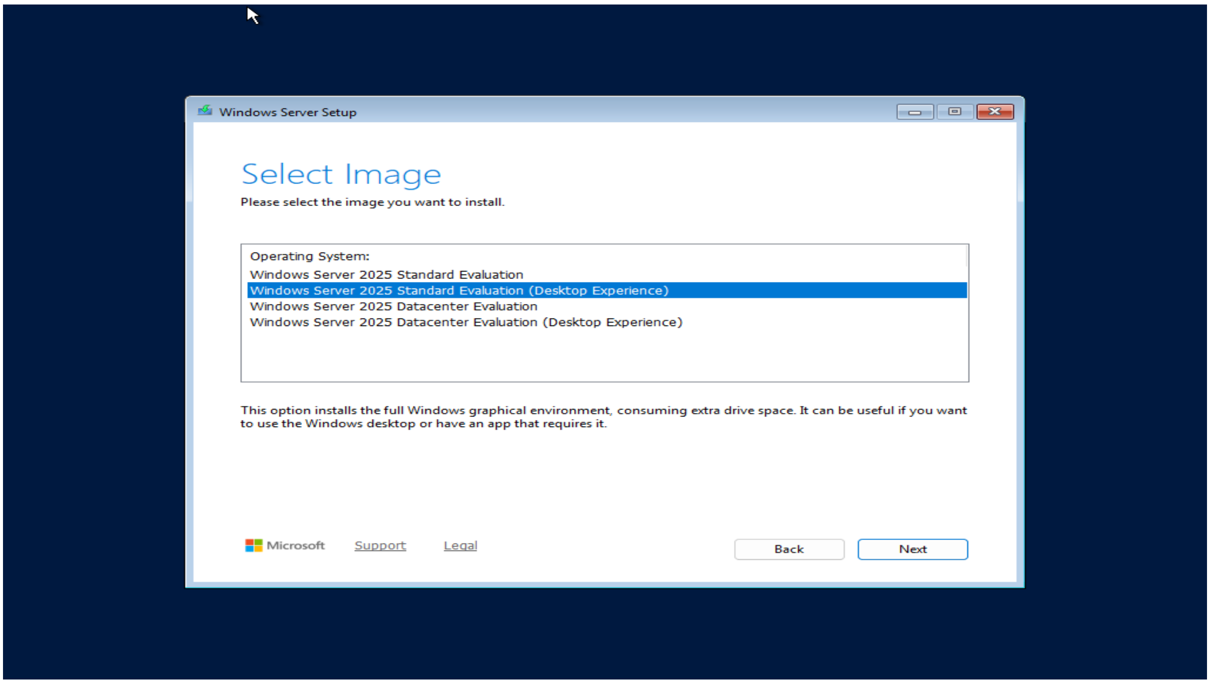
- Language to install
- Time and currency format
- Keyboard/input method





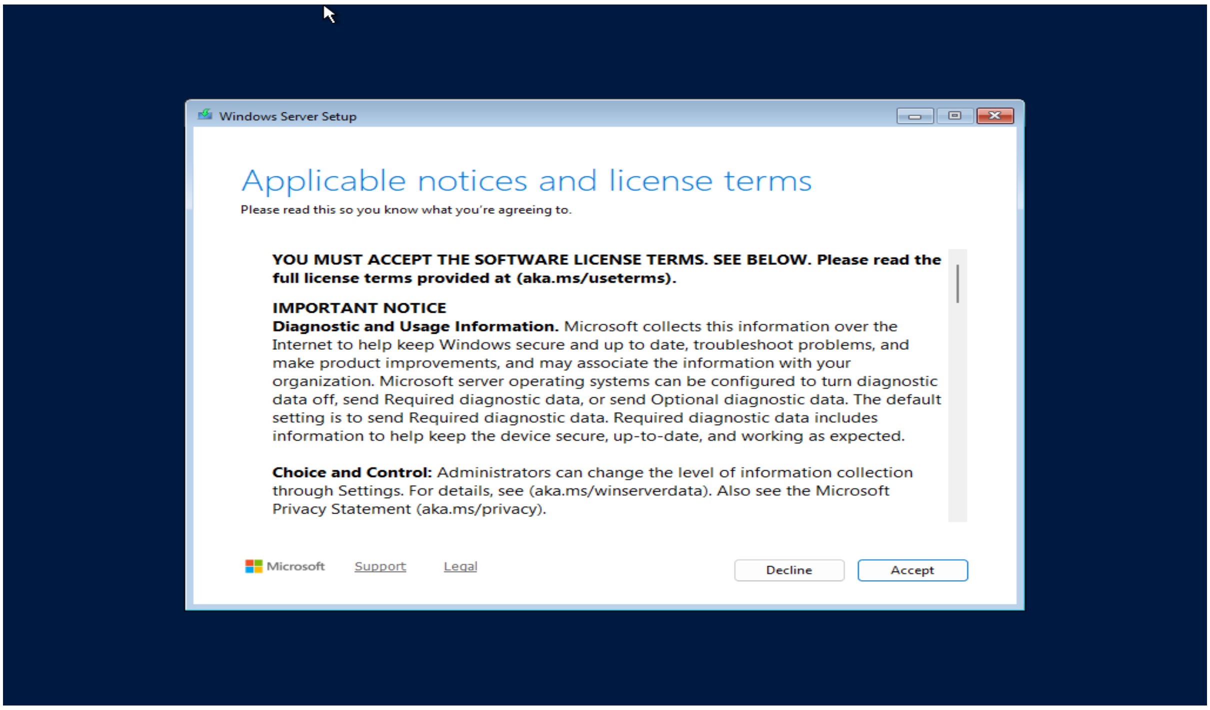
Step 4 — Select the Windows Server Edition

You will see several Windows Server editions.
For an evaluation installation, choose the edition specified by your instructor.
If you need the graphical interface, choose an option ending in:



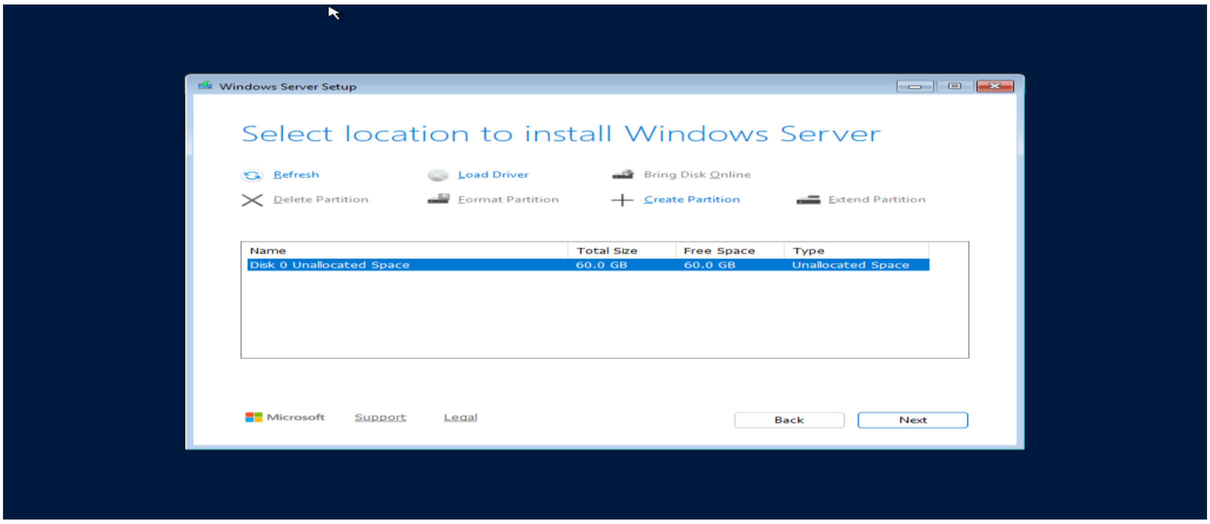
Step 5 — Accept the License Terms

Read the license terms and select:
I accept the Microsoft Software License Terms
Then click Next.



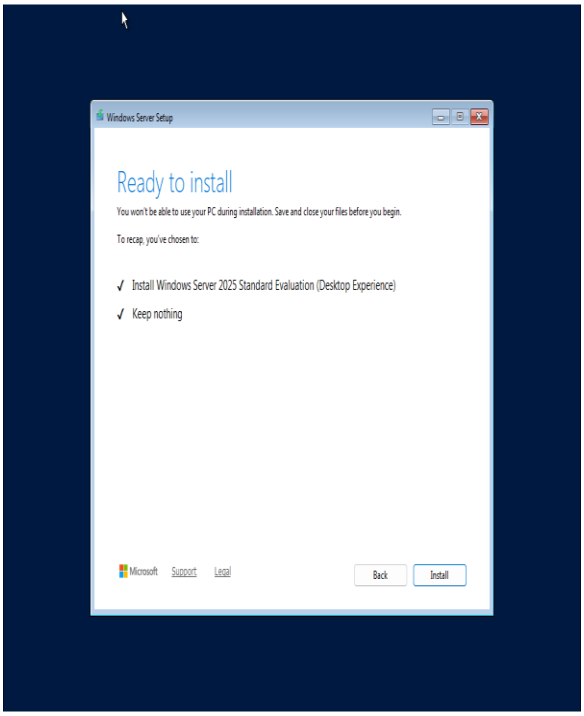
Step 8 — Select the Virtual Disk

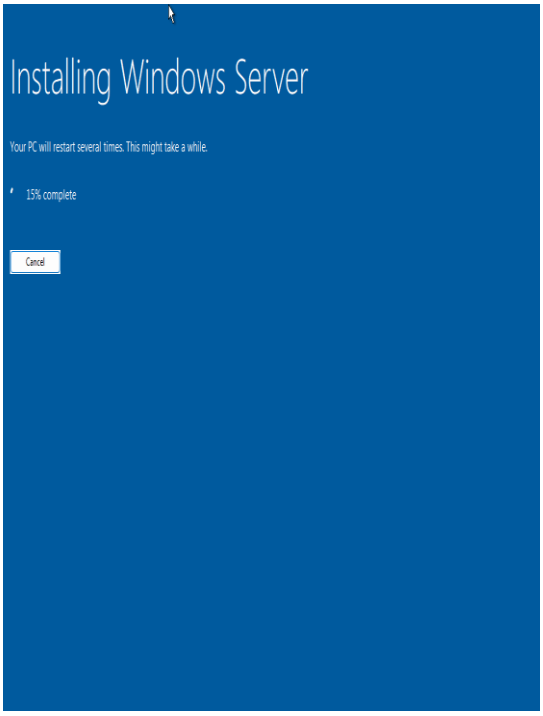
Windows Setup should display the virtual hard disk you created.
Select the available unallocated space and click:
Next
Windows Setup will automatically create the required partitions.



Step 9 — Wait for Installation

Windows will copy files and install the operating system.
The VM may restart several times.

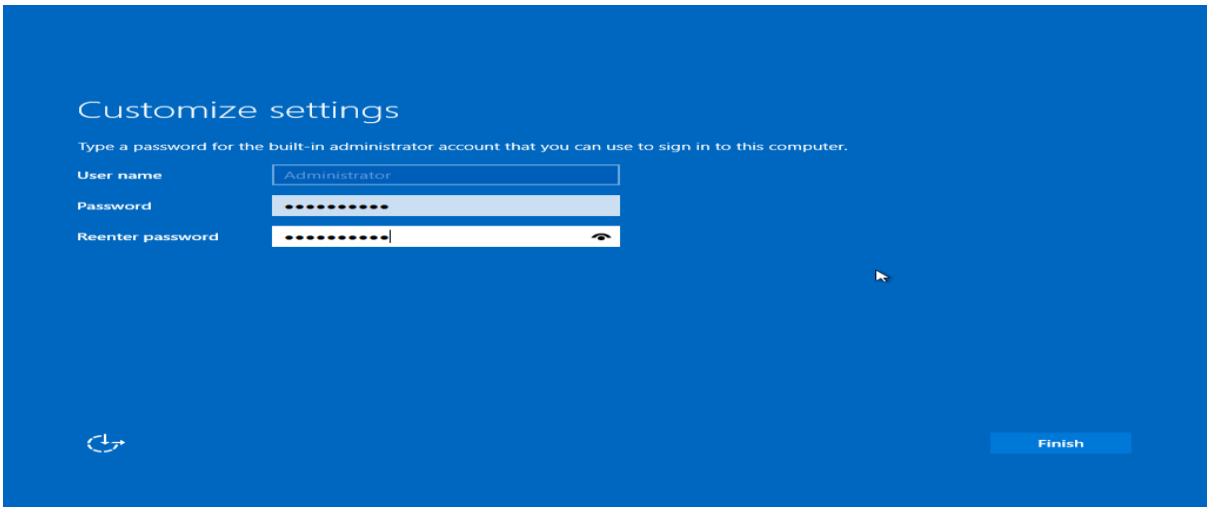




Step 10 — Set the Administrator Password

After installation, Windows Server will ask you to create the password for the built-in Administrator account.

Create a strong password.



Step 11 — Log In

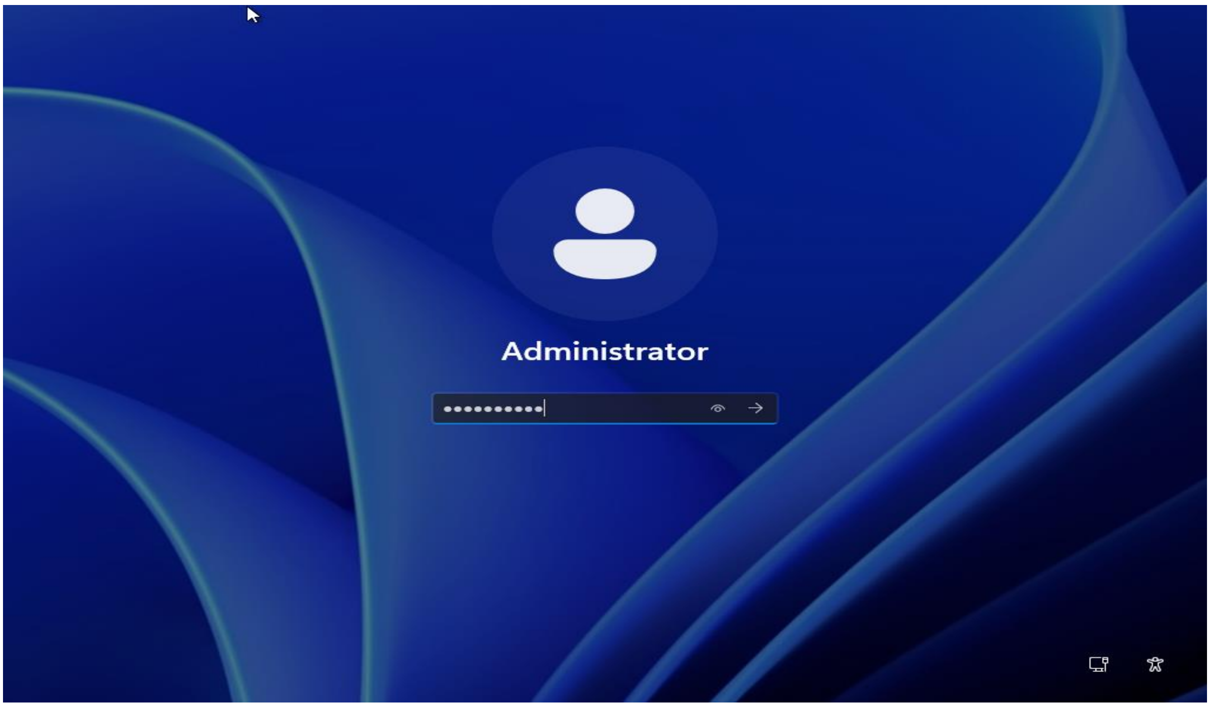
Press:

Ctrl + Alt + Delete

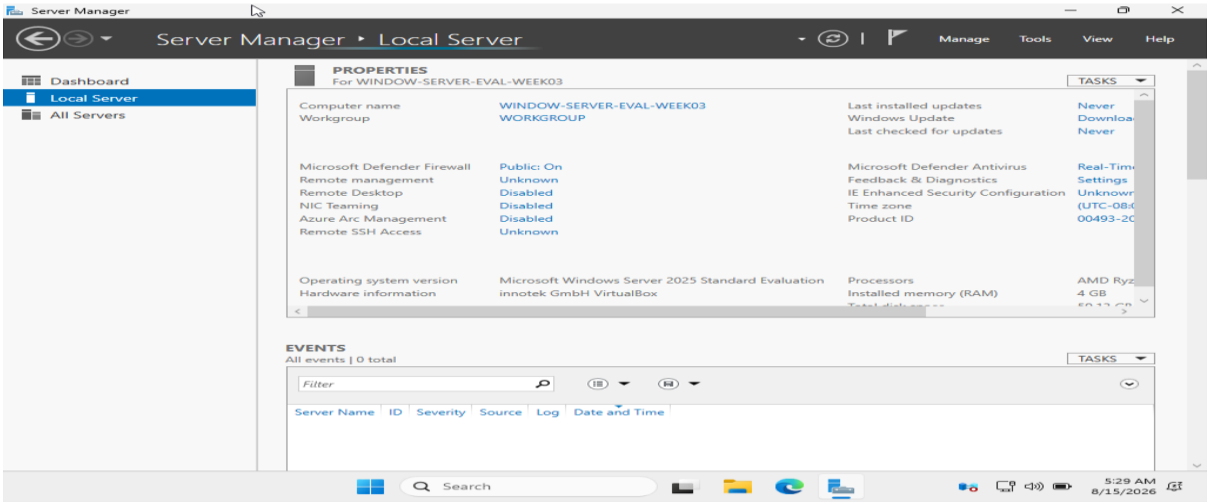
Then log in using:

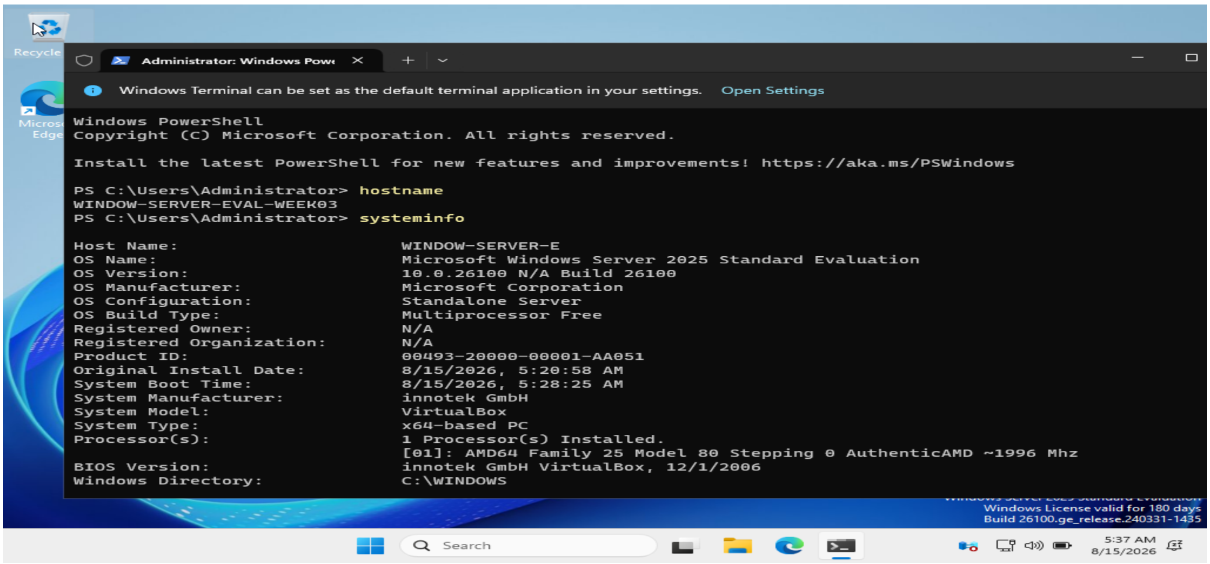
Administrator

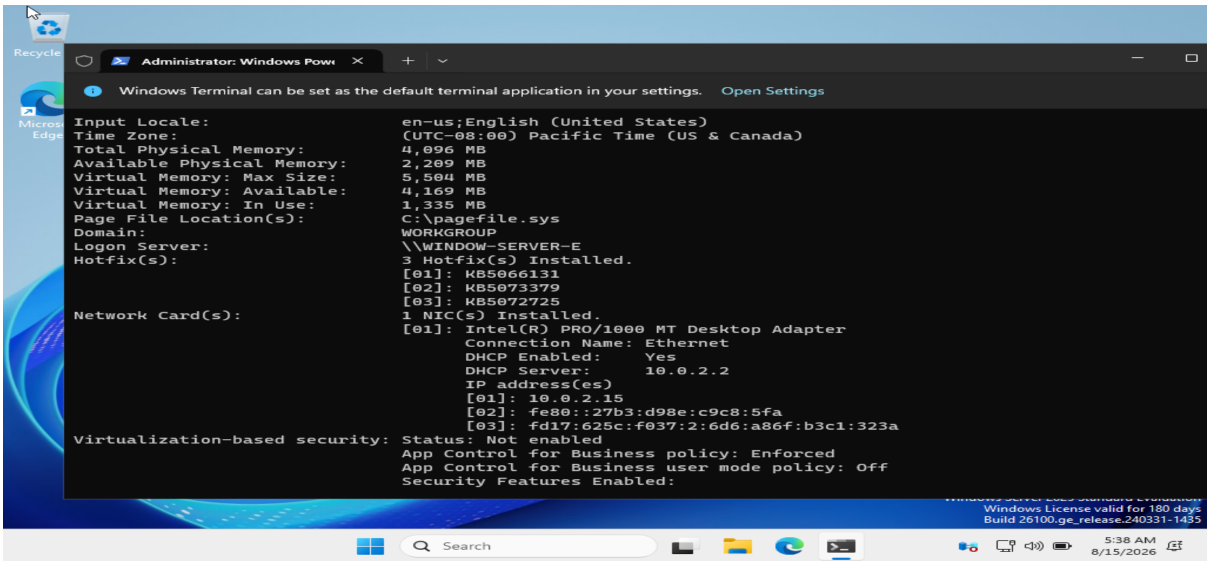
and the password you created.



Configuration Details







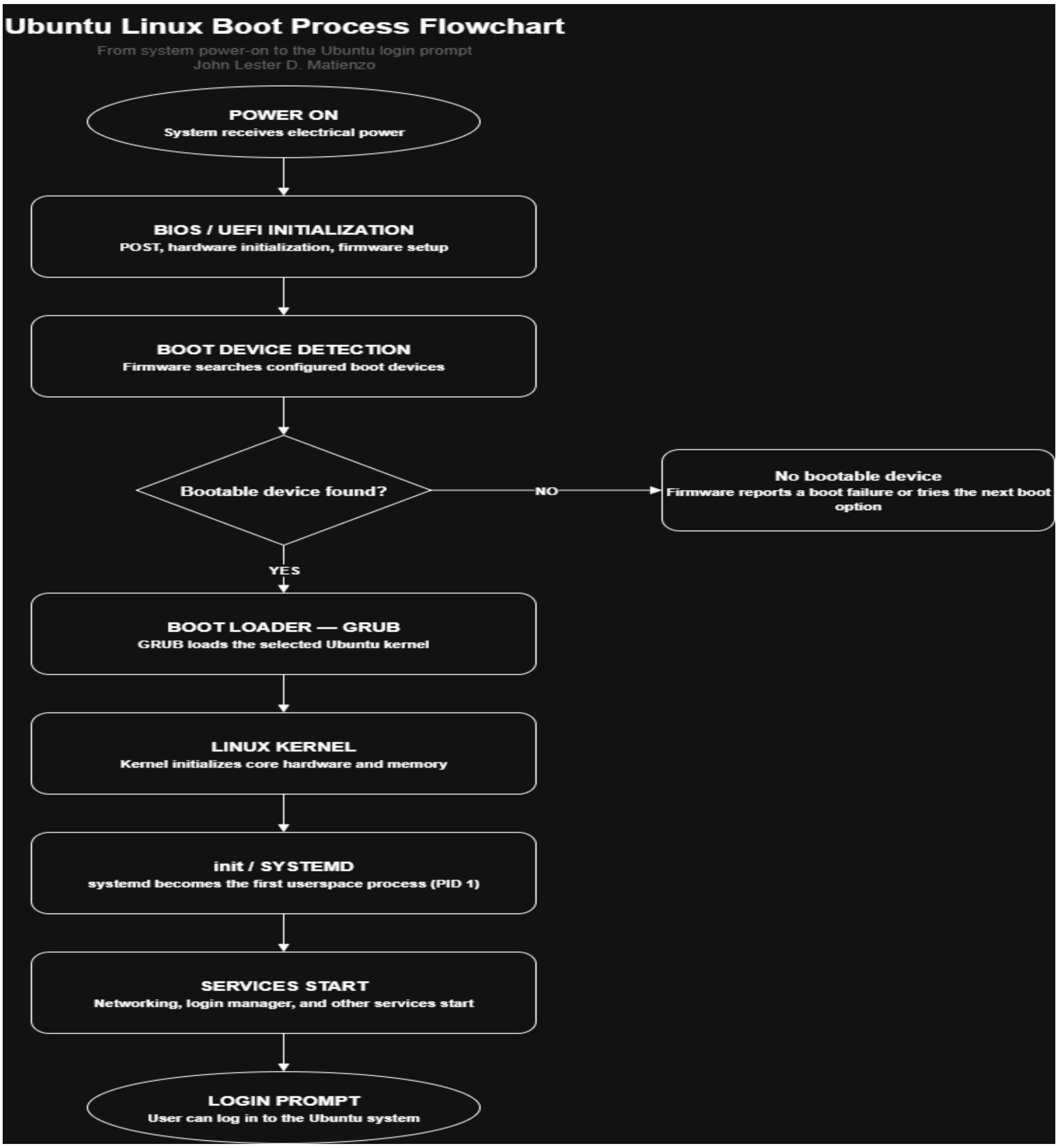
Verification Commands

Command	Purpose
hostname	Displays the computer/host name
ipconfig	Displays IP address information
ipconfig /all	Displays detailed network information
systeminfo	Displays detailed system information
whoami	Displays the currently logged-in account
winver	Displays Windows version information
ping 8.8.8.8	Tests network connectivity

BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is traditional firmware that initializes hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is the modern firmware interface designed to replace legacy BIOS.
Boot Process	Performs POST, looks for a bootable device, reads boot information from the MBR, and starts the bootloader.	Initializes hardware and loads a UEFI bootloader from the EFI System Partition. It can also provide a more flexible boot manager. (UEFI Forum)
Maximum Disk Support	Traditionally limited by the MBR partitioning scheme to about 2 TB for standard 512-byte sectors.	Works with GPT and supports disks much larger than 2 TB. GPT theoretically supports capacities up to about 9.4 ZB under the commonly referenced 512-byte-sector calculation. (Microsoft Learn)
Partition Style	Mainly uses MBR (Master Boot Record).	Commonly uses GPT (GUID Partition Table). GPT supports many more partitions and maintains backup partition information. (UEFI Forum)
Security Features	Has limited built-in boot security compared with modern UEFI systems.	Supports Secure Boot, which can verify digitally signed boot software before allowing it to run. (Microsoft Support)
Speed	Generally, has a slower and more limited startup process.	Generally, provides faster boot and resume times because of its newer firmware architecture. (Microsoft Learn)
Advantages	Simple, widely compatible with older hardware, and useful for legacy systems.	Faster startup, better security, support for large drives, GPT, more partitions, and a more flexible firmware environment.
Disadvantages	Older technology, limited disk/partition support, fewer security features, and less flexibility.	More complex than traditional BIOS and may have compatibility issues with very old operating systems or hardware.
Modern Usage	Mostly found in older computers and legacy configurations.	The standard choice for modern computers and operating systems. Microsoft recommends using UEFI mode when possible because it provides additional security features. (Microsoft Learn)

Boot Process Flowchart



Windows Server Installation Summary

The Windows Server Evaluation operating system was successfully installed inside the Oracle VM VirtualBox using the Windows Server Evaluation ISO. The virtual machine was configured with the required CPU, memory, storage, and network resources before beginning the installation. During the installation process, the appropriate Windows Server edition was selected, the license agreement was accepted, the virtual disk was configured, and an Administrator password was created. After the installation was completed, the server was started successfully and basic system and network information was verified using Windows commands.

OS Comparison Table

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary commercial software. Evaluation editions can be used for testing, while production use requires appropriate Microsoft licensing.	Open-source Linux distribution based on Debian. Available without a traditional per-server license fee.	Open-source enterprise Linux distribution designed to be compatible with Red Hat Enterprise Linux.
User Interface	Provides a graphical interface through Desktop Experience. Server Manager and other Windows administration tools are available.	Primarily command-line based. A graphical interface can be installed, but servers commonly operate without one.	Primarily command-line based. Graphical tools can be installed, when necessary, but enterprise servers commonly use CLI administration.
Package Management	Uses Windows-specific package and software management tools. Administrators commonly use Windows Update, Microsoft package tools, PowerShell, and other management utilities.	Uses APT and Debian packages. Common commands include apt update, apt install, and apt upgrade.	Uses DNF and RPM packages. Common commands include dnf install, dnf update, and dnf remove.
Security	Includes Windows Defender, Windows Firewall, access controls, Active Directory security features, security policies, and Microsoft security updates.	Uses Linux permissions, firewall tools such as UFW, AppArmor, SSH security, and regular security updates.	Uses Linux permissions, SELinux, firewalld, SSH security, and enterprise-oriented security controls.
Performance	Can require more system resources, particularly when using a graphical interface. It is optimized for Microsoft enterprise environments.	Generally lightweight and efficient, especially when operated without a GUI. Suitable for servers with limited resources.	Designed for stability and enterprise workloads. Can perform efficiently when configured without unnecessary services.
Typical Enterprise Use Cases	Active Directory, DNS, DHCP, file and print services, Windows-based applications, Microsoft SQL Server environments, and enterprise management.	Web servers, cloud infrastructure, application servers, containers, development environments, databases, and network services.	Enterprise servers, web applications, databases, virtualization, cloud infrastructure, and environments requiring RHEL compatibility.
Advantages	User-friendly GUI, strong Microsoft ecosystem integration, Active Directory, extensive enterprise management tools, and broad compatibility with Windows applications.	Free and open source, large community, extensive software repositories, efficient resource usage, and strong cloud/server support.	Free and open source, stable, enterprise-focused, RHEL-compatible, predictable lifecycle, and strong security features.
Disadvantages	Licensing can be expensive, may require more resources, and has greater dependence on the Microsoft ecosystem.	Administration often requires command-line knowledge, and some commercial Windows applications are not natively supported.	Can have a steeper learning curve for beginners and may require more Linux administration knowledge.

Problems Encountered

During the installation process, several minor challenges were encountered. The first was the long download time of the Windows Server Evaluation ISO. Since the ISO file is relatively large, the download depended on the available internet speed and took some time to complete. This delayed the installation because the ISO had to be completely downloaded before it could be attached to the virtual machine.

Another challenge was identifying the appropriate Windows Server installation option because multiple editions and installation options were presented during setup. Careful checking was required to select the appropriate Evaluation edition with Desktop Experience.

The virtual machine configuration also required checking before starting the installation, particularly the storage and ISO settings, to ensure that the VM would correctly boot from the Windows Server installation media.

Solutions Implemented

To address the problems encountered during the installation process, here's the solutions I implemented. For the long Windows Server Evaluation ISO download, I allowed the download to complete before proceeding with the virtual machine installation. I also checked that the ISO file was completely downloaded and available on the computer before attaching it to the VM. This prevented installation errors that could occur from using an incomplete or corrupted installation file.

For the difficulty in identifying the appropriate Windows Server edition, I carefully reviewed the available installation options and selected the required Evaluation edition with Desktop Experience. This allowed me to continue with a graphical server environment that was appropriate for the activity.

Before starting the installation, I also reviewed the virtual machine settings, including the storage, processor, memory, and network configuration. Checking these settings beforehand helped ensure that the virtual machine had the necessary resources and that the Windows Server ISO was correctly connected as the boot media.

Lessons Learned

Through this activity, I learned that installing a server operating system like Window Server Evaluation requires proper preparation before starting the installation. I learned how to configure a virtual machine by assigning appropriate memory, processor resources, storage, and network settings. I also gained a better understanding of the Windows Server installation process, including selecting the correct server edition, configuring the administrator account, and checking the installed system.

I also learned the importance of verification after installation. Commands such as hostname, ipconfig, systeminfo, and whoami provide useful information for confirming that the server is working correctly. The activity also helped me understand the difference between BIOS and UEFI and how firmware is involved in the computer boot process.

Overall, the activity improved my practical skills in system administration because I was able to perform the installation, troubleshoot issues, and document the process instead of only studying the concepts theoretically.

References

Microsoft. (n.d.). *Windows Server documentation*. Microsoft Learn. https://learn.microsoft.com/en-us/windows-server/

Microsoft. (n.d.). *Microsoft Evaluation Center*. Microsoft. https://www.microsoft.com/en-us/evalcenter/

Microsoft. (n.d.). *Windows Server 2025 Trial*. Microsoft. https://info.microsoft.com/ww-landing-evaluate-windows-server-2025.html

Oracle. (n.d.). *Oracle VirtualBox documentation*. Oracle. https://docs.oracle.com/en/virtualization/virtualbox/

Oracle. (2024). *Oracle VM VirtualBox User Guide*. Oracle. https://docs.oracle.com/en/virtualization/virtualbox/7.0/user/index.html

Microsoft. (n.d.). *Boot to UEFI mode or legacy BIOS mode*. Microsoft Learn. https://learn.microsoft.com/en-us/windows-hardware/manufacture/desktop/boot-to-uefi-mode-or-legacy-bios-mode